

NORMAL DISTRIBUTION AND STANDARD DEVIATION

Q1. Generate 1000 Random Numbers Following a Normal Distribution and Find Mean, Median and Mode

Definition

A **normal distribution** is a continuous probability distribution that has a symmetric, bell-shaped curve. In a normal distribution, the mean, median, and mode are approximately equal.

For a **standard normal distribution**:

- Mean (μ) = 0
- Standard deviation (σ) = 1
- Total observations (n) = 1000

Given

Parameter	Value
Number of observations	1000
Mean (μ)	0
Standard deviation (σ)	1

Measures of Central Tendency

Mean:

The mean represents the average value of the data.

$$\mu = 0$$

Median:

The median is the middle value when all observations are arranged in ascending or descending order.

For a standard normal distribution:

$$\text{Median} \approx 0$$

Mode:

The mode is the value that occurs most frequently.

For a perfectly normal distribution:

$$\text{Mode} \approx 0$$

Result

$$\text{Mean} \approx \text{Median} \approx \text{Mode} \approx 0 \quad \boxed{\text{Mean} \approx \text{Median} \approx \text{Mode} \approx 0}$$

Conclusion

The generated data follows a **standard normal distribution**, where the center of the distribution is approximately 0 and the standard deviation is 1.

Q2. Bell Curve of a Normal Distribution

Definition

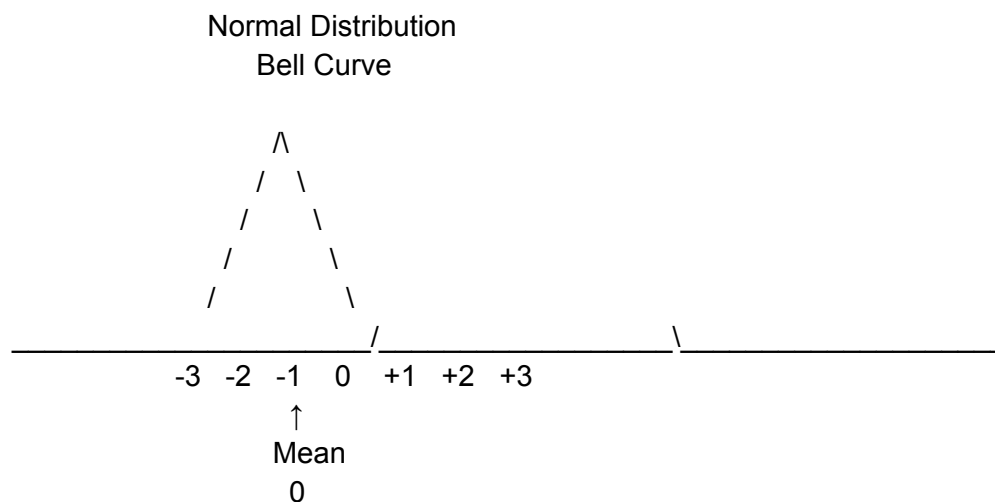
A **bell curve** is the graphical representation of a normal distribution. It is symmetric around the mean.

The center of the curve represents the **mean** (μ).

For a standard normal distribution:

$$\mu = 0 \quad \sigma = 1$$

Bell Curve Diagram



Important Points

- The curve is **symmetric** around the mean.
- Mean = Median = Mode in an ideal normal distribution.
- Most observations are concentrated around the mean.
- As we move away from the mean, the number of observations decreases.

Mean

The vertical line at:

$$\mu = 0$$

represents the mean of the distribution.

Q3. Percentage of Data Within 1, 2 and 3 Standard Deviations

Definition

Standard deviation (SD) measures how much the data values are spread around the mean.

For a normal distribution, the **Empirical Rule** states:

- Approximately **68%** of data lies within 1 standard deviation.
- Approximately **95%** of data lies within 2 standard deviations.
- Approximately **99.7%** of data lies within 3 standard deviations.

Given

$$\mu = 0 \quad \sigma = 1$$

1 Standard Deviation

Interval:

$$\mu - \sigma \leq X \leq \mu + \sigma$$

Substituting the values:

$$0 - 1 \leq X \leq 0 + 1 \quad -1 \leq X \leq 1$$

Percentage:

$$68\%$$

For 1000 observations:

$$1000 \times \frac{68}{100} = 680$$

Therefore, approximately:

680 observations

2 Standard Deviations

Interval:

$$\mu - 2\sigma \leq X \leq \mu + 2\sigma \quad -2 \leq X - \mu \leq +2$$

Percentage:

95%

For 1000 observations:

$$1000 \times \frac{95}{100} = 950$$

Therefore, approximately:

950 observations

3 Standard Deviations

Interval:

$$\mu - 3\sigma \leq X \leq \mu + 3\sigma \quad -3 \leq X - \mu \leq +3$$

Percentage:

99.7%

For 1000 observations:

$$1000 \times \frac{99.7}{100} = 997$$

Therefore, approximately:

997 observations

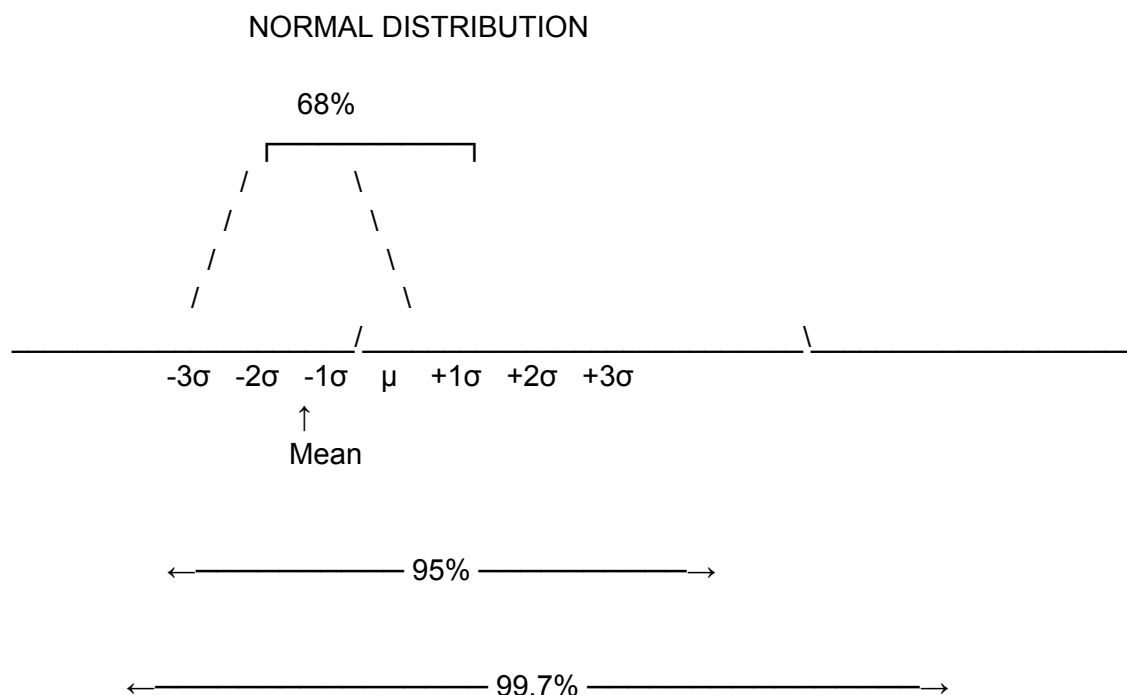
Summary Table

Standard Deviation	Range	Percentage	Approx. Observations out of 1000
1 SD	-1 to +1	68%	680
2 SD	-2 to +2	95%	950
3 SD	-3 to +3	99.7%	997

Q4. Bell Curve with 68%, 95% and 99.7% Intervals

Empirical Rule

The normal distribution can be divided according to standard deviations from the mean.



Interpretation

Within 1 Standard Deviation:

$$\mu \pm 1\sigma = 68\% \quad \boxed{\mu \pm 1\sigma = 68\%}$$

This means approximately 68% of the observations are close to the mean.

Within 2 Standard Deviations:

$$\mu \pm 2\sigma = 95\% \boxed{\mu \pm 2\sigma = 95\%}$$

This means approximately 95% of the observations fall within this range.

Within 3 Standard Deviations:

$$\mu \pm 3\sigma = 99.7\% \boxed{\mu \pm 3\sigma = 99.7\%}$$

This means almost all observations fall within three standard deviations of the mean.